special language for smart contract implementation. To summarize, Solidity is a custom language designed for smart contract development.

Let’s explore some of the features of the Solidity language before coding the Counter smart contract.

2.3.1 Solidity language

Solidity is an object-oriented, high-level language for implementing smart contracts, influenced by C++, Python, and JavaScript. Solidity is statically typed and supports inheritance, libraries, and user-defined types; it also provides many useful features for developing blockchain applications. Because its syntax and semantics are similar to those of languages you may know, I will not explicitly discuss the language elements of Solidity but will introduce and explain them as we go, using code snippets. You will use the Remix integrated development environment for writing, editing, compiling, deploying, and testing the code on a simulated blockchain.

2.3.2 Smart contract code for Counter

In this section, you’ll develop the code for the smart contract specified in the design diagrams of figure 2.5. The complete code in Solidity is shown in listing 2.1. The first line specifies the version of the language used for this code. It’s mandatory to include this directive to ensure that you use the compiler version that matches the Solidity language version used in the code. In this case, you’ll be using version